

Karneeshkar V

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[Portfolio](#) | [Terminal portfolio](#)

PROFESSIONAL SUMMARY

AI engineer focused on building, fine-tuning, and shipping LLM-powered products. Hands-on with LLaMA 3.1 fine-tuning, agentic workflows, retrieval, and MLOps; strong backend foundation in Python/FastAPI and cloud deployment. Track record of shipping measurable impact—e.g., 88% intent classification in production chatbots and 94% vision accuracy on edge devices—and leading cross-functional teams to reliable releases.

TECHNICAL SKILLS

Languages: Python, Bash, Go, SQL, C/C++

AI & Data: PyTorch, TensorFlow, Computer Vision, Sensor Data Processing

Web & Cloud: FastAPI, Node.js, REST APIs, llama.cpp, AWS, Docker, Git

Databases: PostgreSQL, SQLite, MongoDB

Soft skills: Communication, Teamwork, Active listening, Problem solving, Critical thinking, Time management

EXPERIENCE

Software Developer 2

Jul 2025 – Current

2Cents Capital

Bengaluru, India

- Led and scaled cross-functional Mobile and AI teams (10+ engineers) to deliver high-impact features with 95% on-time release rate.
- Published a fully responsive investment application on both iOS App Store and Google Play, achieving <0.5s cold-start time.
- Designed and deployed AI agents across the end-to-end customer journey—including customer support, onboarding, and market research—improving support resolution speed by 40% and reducing manual intervention costs by 30%.

Software Developer

Jan 2025 – Jun 2025

Visteon

Chennai, India

- Created diagnostic routines for JLR platforms that detect system anomalies with 96% accuracy, boosting vehicle reliability.
- Optimized real-time dashboard software (SomeIP on QNX/Linux) in C, reducing processing latency by 25%.
- Wrote unit and functional tests to reach 99.8% code coverage across the code-base with zero deviations.

Automation Intern

Dec 2024 – Jan 2025

Procter & Gamble

Hyderabad, India

- Implemented IIoT monitoring on five production lines, ingesting 2M+ data points daily with 99.9% uptime.
- Automated quality-control workflows, cutting manual inspection time 40% and improving defect detection 15%.

IoT & LoRaWAN Developer

May 2024 – Jun 2024

Tynatech Ingenious Pvt Ltd

Noida, India

- Built RESTful APIs / webhooks connecting 50+ LoRaWAN devices to cloud platforms.
- Optimized UART-LoRaWAN interface to raise throughput 30% via efficient buffer management.
- Added MQTT streaming for real-time gateway visualization, handling 100+ msgs s⁻¹.

Web Development Intern

Oct 2023 – Dec 2023

Intellect Design Arena

Chennai, India

- Developed PHP-based financial portals with 30% faster page loads.
- Managed cloud infra for five web apps, configuring auto-scaling for 99.9% availability.
- Created analytics dashboards delivering real-time engagement and conversion insights.

R&D Intern

Aug 2023 – Sep 2023

Hindustan Aeronautics Ltd (HAL)

Bangalore, India

- Optimised autopilot algorithms, cutting computational overhead 22% while meeting safety constraints.
- Authored C++ simulation tool covering 200+ control-system test scenarios.
- Assisted avionics teams with system documentation and verification protocols.

PROJECTS

- Full-Stack AI Personal Assistant** | *Go, Google A2A, PostgreSQL, React* Dec 2024
- Engineered and deployed a cloud assistant that centralizes and automates personal workflows.
 - Built asynchronous agents to orchestrate complex Google Workspace operations (mail, calendar, docs).
 - Modeled a personal knowledge graph to ground answers and personalize actions.
- LLaMA 3.1 Fine-Tuning for Code Reasoning** | *Python, RL (GPRO), Hugging Face* Apr 2025
- Fine-tuned LLaMA-3.1 8B on a paired-coding dataset (link) using a single H100 to boost code reasoning.
 - Applied GPRO reinforcement learning with Gemma 3 as a reward model (inspired by DeepSeek-R1).
 - Improved pass@1 by ~15% on a custom eval and averaged +10% on public coding benchmarks.
- Instagram Automation Bot** | *Python, FastAPI, Redis, AWS* Dec 2024
- Deployed cloud bot handling 1000+ daily interactions with 99.5% uptime.
 - Added LLM-based sentiment/intent detection achieving 88% accuracy in production.
 - Introduced Redis caching to cut Meta Graph API calls 65%; p50 response < 500 ms.
- Multipurpose VR Gloves with Gesture Recognition** | *Unity, C#, ESP32, ML* Sep 2024
- Built VR gloves capturing 15+ gestures at 92% accuracy using flex-sensor ML pipelines.
 - Developed web UI for mapping gestures to 20+ system commands; integrated Unity for gaming/assistive control.
- Embedded Athlete-Performance Monitoring System** | *ESP32, PyTorch, OpenCV* Mar 2024
- Ran edge CV on ESP32 to analyze runner form in real time; fused MPU6050, VO₂, and temperature signals.
 - Built a decision-tree classifier with a Flask/Plotly dashboard for actionable feedback.
- Smart Rhesus Macaque Deterrent System** | *Jetson Nano, CV, Ultrasonic* Mar 2024
- Achieved 93–94% macaque detection across varied lighting; ultrasonic emitters cut intrusions 90%.
 - Designed a robust CV pipeline (tracking + classification) optimized for low-power edge deployment.

EDUCATION

Vellore Institute of Technology Chennai, India
B.Tech. Electronics & Computer Engineering; GPA: 8.66/10